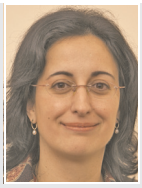
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Underground carbon dioxide storage in saline formations

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This paper reviews key areas of carbon dioxide storage in saline aquifers. Among several potential geological carbon dioxide storage sites, saline aquifers offer the highest storage potential capacity and are relatively common worldwide. Geological storage of carbon dioxide in saline formations is achieved through various physical and chemical trapping mechanisms. When carbon dioxide is injected in the subsurface, it is first trapped by primary trapping mechanisms, which are static and hydrodynamic trapping below the cap rock. The secondary trapping mechanisms, solubility, mineral and residual trapping, are much slower and have a much longer timeframe for operating than primary trapping mechanisms. Knowledge regarding storage capacity in saline formations is highly uncertain due to their poor characterisation, heterogeneity and trapping mechanisms operating at different timeframe scales, which pose a huge challenge to establish a reliable methodology for storage capacity estimates. The potential cost of storage in saline aquifers is reasonably well known. However, the lack of any economic benefits makes this storage option less economically attractive than other geological sequestration strategies.

1. INTRODUCTION

In 2004, carbon dioxide represented 77% of the total world anthropogenic greenhouse gas (GHG) emissions. Future predictions show a substantial increase in carbon dioxide emissions throughout this century in the absence of specific actions for stabilisation of atmospheric GHG concentrations. This is largely due to a continued and increased use of cheap, available and globally distributed fossil fuels, mainly coal, especially as economies and standards of living in the developing world increase (IPCC, 2005).

Carbon capture and storage (CCS) has the potential to reduce overall mitigation costs and maintain the use of coal as an energy source in the short term while future non-fossil-based fuel energy systems are developed and implemented (IPCC, 2005; Lackner *et al.*, 1998; Mazzotti *et al.*, 2009). CCS is a process consisting of the separation of carbon dioxide from industrial and energy-related sources, its transport to a storage location and long-term isolation from the atmosphere (IPCC, 2005). Available evidence suggests that, worldwide, there is a technical potential of at least 2000 $GtCO_2$ of storage capacity in geological formations, set against carbon dioxide emissions from fossil-fuel use in the year 2000 of 23, 5 $GtCO_2$ (IPCC, 2005). Different target formations for carbon dioxide injection include depleted oil and gas reservoirs, unminable coal seams and deep saline reservoirs (IEAGHG, 2010; Juanes *et al.*, 2006); their storage capacities are presented in Table 1.

Oil and gas reservoirs consist of porous rocks covered by impermeable cap rock, which is often dome shaped. Following more than a century of intensive petroleum exploitation, thousands of oil and gas fields are approaching the ends of their economically productive lives. Some of these depleted fields could act as effective storage sites for carbon dioxide. In most oil fields, only a portion of the original oil in place is recovered using standard petroleum extraction methods. Carbon dioxide injected into suitable, depleted oil reservoirs could enhance oil recovery by typically 10–15% of the original oil in place in the reservoir. This is an established technique, called carbon-dioxide-enhanced oil recovery (CO_2-EOR). The additional oil production could more than offset the cost of carbon dioxide capture and injection (IEAGHG, 2010). The storage potential of oil and gas fields is 675–900 $GtCO_2$ (IPCC, 2005).

Carbon-dioxide-enhanced coal bed methane (CO_2-ECBM) production is another potential underground storage option as the carbon dioxide remains stored within the seam, locking it up in a secure and stable way providing the coal is never mined (EAGHG, 2010). Carbon dioxide has a greater affinity for coal than methane (CH_4) or nitrogen (N_2) (Bachu, 2008; Mazzotti *et al.*, 2009) and therefore carbon dioxide injected into coal seams preferentially displaces methane that exists in the coal, which might then be recovered as free gas. Injection of carbon dioxide enhances primary coal bed methane recovery, which is typically limited to only 20–60%, while at the same time sequestering carbon dioxide (Mazzotti *et al.*, 2009). In addition, the sale of the methane produced allows, in principle, the offsetting of costs associated with CCS operation, as is the case for EOR (Mazzotti *et al.*, 2009). Unminable coal seams are estimated to have a smaller, but still significant, potential for carbon dioxide storage than other geological formations of 3–200 $GtCO_2$ (IPCC, 2005; Mazzotti *et al.*, 2009).

While injection into hydrocarbon reservoirs and coal seams has benefits associated with enhanced oil and gas production, storage in deep saline aquifers has the greatest storage potential (Table 1) (Obi and Blunt, 2006). It is estimated that up to 10 000 $GtCO_2$ could be stored worldwide in aquifers (IPCC,

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Reservoir type	Storage capacity: $GtCO_2$	
	Lower estimate	Upper estimate
Oil and gas fields	675*	900*
Unminable coal seams	3–15	200
Deep saline formations	1000	Uncertain, but possibly 10^4

* These numbers would increase by 25% if 'undiscovered' oil and gas fields were included in this assessment

Table 1. Storage capacity for several geological storage options (storage capacity includes storage options that are not economical) (IPCC, 2005)

2005), which is equivalent to many centuries of carbon dioxide emissions at the current rate of around 25 Gt/year (Obi and Blunt, 2006). Aquifers can be divided into two groups

(*a*) freshwater aquifers (with less than about 1000–3000 mg/l of dissolved solids), which are sources of drinking water
(*b*) saline aquifers, which are the most suitable for disposal of carbon dioxide.

Saline formations are deep sedimentary rocks saturated with formation waters or brines containing high concentrations of dissolved salts (IPCC, 2005). The most suitable reservoirs are those at depths greater than 800 m, as, at this depth, the carbon dioxide will behave more like a liquid than a gas, enabling more carbon dioxide to be stored. They contain saline water and are unsuitable for supplying potable water. Aquifers with salinities exceeding 10 000 mg/l total dissolved solids (TDS) are excluded by the US Environmental Protection Agency as underground sources of drinking water (Xu *et al.*, 2004). The injection of carbon dioxide into saline formations is a promising carbon dioxide sequestration option. First, the technology for injecting carbon dioxide into exhausted hydrocarbon reservoirs and deep saline aquifers already exists. Second, even though no direct economic benefits are derived, it should be noted that the carbon dioxide sequestration potential associated with the injection of carbon dioxide into exhausted oil and natural gas reservoirs and deep saline aquifers greatly exceeds that of EOR or ECBM (Bock *et al.*, 2003). Exhausted hydrocarbon reservoirs have the advantage over aquifers in that these fields are proven long-term traps, their geology is well characterised, and their existing surface and subsurface infrastructures could readily be converted for carbon dioxide distribution and injection. On the downside, the locating and sealing of abandoned oil and gas wells could present a challenge, which is not the case for aquifers (Bock *et al.*, 2003). Moreover, aquifers are widely distributed, have the largest volumes of pore space available and are generally located closer to carbon dioxide emission sources (Bock *et al.*, 2003; Hitchon *et al.*, 1999). Saline formations are thus the main interest of this paper.

Some of the risks associated with carbon dioxide capture and geological storage are similar to, and comparable with, any other industrial activity for which extensive safety and regulatory frameworks are in place. Specific risks associated with carbon dioxide storage relate to the operational (injection) phase and post-operational phase, of which the risks of most concern are those posed by the potential for acute or chronic carbon dioxide leakage from the storage site. Hence, local risks to health and safety, environment and equity need to be properly assessed and managed (Bachu, 2008).

2. CARBON DIOXIDE TRAPPING MECHANISMS IN SALINE FORMATIONS

One of the most promising options for reducing the atmospheric concentration of anthropogenic carbon dioxide is to inject and store this GHG in underground formations such as deep saline aquifers. This has already been accomplished in Sleipner West field and Snøhvit field, both in Norway (Bickle *et al.*, 2007; Bock *et al.*, 2003; Chiquet *et al.*, 2007; Laate, 2008).

If carbon dioxide is injected into suitable saline formations at depths below 800 m, various physical and geochemical trapping mechanisms should prevent it migrating to the surface. Geological storage of carbon dioxide is achieved through a combination of physical and chemical trapping mechanisms that are effective over different timescales (IPCC, 2005) but may operate simultaneously (Bachu *et al.*, 2007). The most effective storage sites are those where carbon dioxide is immobile because it is trapped permanently under a thick, low-permeability seal or is converted to solid minerals or undergoes a combination of physical and chemical trapping mechanisms (IPCC, 2005).

Trapping mechanisms can be divided into primary and secondary processes (Table 2). When carbon dioxide is injected in the subsurface, it is first trapped by primary trapping mechanisms (which are static) and hydrodynamic trapping below the cap rock in deep saline aquifers (Bachu *et al.*, 2007). The secondary processes – solubility, mineral and residual trapping – are much slower and occur over a much longer timeframe than the primary mechanisms (of the order of centuries to millennia). The contribution of these secondary trapping mechanisms is almost negligible during the injection phase of carbon dioxide geological storage but they contribute to increasing storage security in the longer term (Bachu, 2008).

Trapping mechanisms can be also distinguished in terms of physical and geochemical processes (Table 2). Over time, the contribution of each trapping mechanism changes. Initially, physical trapping is the dominant mechanism for retaining carbon dioxide in aquifers. As carbon dioxide migrates through aquifers, residual gas and solubility trapping tend to be significant mechanisms. These secondary trapping mechanisms enhance storage security because the risk of leakage of the products of such trapping mechanisms does not depend directly on the integrity of the cap rock on which physical trapping depends (Suekane *et al.*, 2008).

Static (structural and stratigraphic)	P	Ph
Residual	S	Ph
Hydrodynamic	P	G
Ionic and solubility	S	G
Mineral	S	G
Adsorption*	P	Ph

* Adsorption does not take place in saline formations

Table 2. Trapping mechanisms divided in terms of primary (P)/secondary (s) and physical (Ph)/geochemical (G)

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2.1. Physical trapping mechanisms

Physical trapping occurs when carbon dioxide is immobilised as a free gas or supercritical fluid. There are two types of physical trapping

(*a*) static trapping in stratigraphic and structural traps or in man-made caverns
(*b*) residual gas trapping in the pore space at irreducible gas saturation (Bachu, 2008; Bachu *et al.*, 2007).

2.1.1. Static trapping mechanisms. Initially, static trapping is the primary mechanism for storing carbon dioxide in geological formations. It refers to permeable geological media overlain and surrounded by cap rock, a low-permeability porous rock saturated with brine, whose shape precludes the upward and lateral movement of a buoyant fluid such as carbon dioxide (Bachu, 2008; Chiquet *et al.*, 2007; IPCC, 2005). Cap rock inhibits the flow of a given fluid at a certain temperature, pressure and geochemical condition. The necessary properties for a rock to act as a seal will be different for different fluids; they are determined by capillary forces, the magnitude of which depends on fluid and rock properties (fluid density, fluid viscosity, rock structure, rock wettability) and pore size (capillary forces almost disappear when pore diameter exceeds 0·5 mm) (Chilingar *et al.*, 2005).

Carbon dioxide injected into saline aquifers migrates unevenly upwards until it reaches the cap rock as a result of buoyancy caused by the density differences between carbon dioxide and formation water (30–50%). The cap rock acts as a barrier and causes the buoyant carbon dioxide to migrate laterally along the inclination of the layer, filling any structural or stratigraphic traps it encounters (Flett *et al.*, 2005; Holloway, 2005; IPCC, 2005; Suekane *et al.*, 2008). If the impermeable layer has a closed dome structure, carbon dioxide can accumulate at high saturations below the layer (Suekane *et al.*, 2008). The permeability of a homogeneous cap rock without faults or fractures is dependent mainly on its capillary entry pressure, which is essentially a function of the size of the pore throats connecting the pores within the rock. However, in real situations, it is unlikely that a cap rock that is perfectly homogeneous and impermeable will be found. A cap rock may be discontinuous and may also contain faults or fractures that could cause it to leak (Holloway, 2005; Rutqvist and Tsang, 2002). Cap rocks can be divided into two categories

(*a*) aquicludes – essentially impermeable strata such as thick rock salt layers (salt beds)
(*b*) aquitards – low-permeability strata such as shales and mudstones through which fluids can migrate, albeit extremely slowly (Holloway, 2005). Shales are the most common seals. They are often porous, but because of the fine grain size, have very high capillary forces that inhibit fluid flow (Selley, 1998).

Storing carbon dioxide by static trapping is similar to storing carbon dioxide in depleted oil and gas reservoirs, the only difference being that the trap is initially saturated with water instead of containing hydrocarbons (Bachu *et al.*, 2007) (hydrocarbon traps were described in detail by Selley (1998)). Static trapping takes place in

(*a*) structural traps – these result from crustal movements (Bachu, 2008) and include those formed by folded or fractured rocks (CO2CRC, 2008; IPCC, 2005)
(*b*) stratigraphic traps – these are the result of depositional and/or diagenetic processes (Bachu, 2008)
(*c*) combinations of both structural and stratigraphic traps (CO2CRC, 2008).

Storage safety is closely related to the ability of the cap rock to prevent – or at least seriously hinder – carbon dioxide leakage. Among the possible sources of leakage, carbon dioxide diffusion through a brine-saturated cap rock is an extremely slow process, considered to be significant only over geological timescales. The most dangerous (i.e. rapid) leakage sources are, by far, capillary breakthrough of carbon dioxide and rock fracture, which occur when the pressure build-up in the reservoir due to carbon dioxide injection is excessive. Carbon dioxide capillary breakthrough pressure is an important constraint in designing and planning a carbon dioxide injection and storage project, as it sets a limit on both the injection pressure and the maximum amount of carbon dioxide that can be stored beneath a given cap rock, and therefore on the reservoir storage capacity (Chiquet *et al.*, 2007). Care must be taken not to exceed the allowable overpressure to avoid fracturing the cap rock or reactivating faults (IPCC, 2005). The regulatory agencies usually require that the injection pressure has to be higher than the initial water pressure in the trap in order to achieve carbon dioxide injection, and the predetermined limiting pressure has to be lower than the maximum bottom hole injection pressure in order to avoid rock fracturing. The maximum injection pressure should also be less than the threshold entry pressure of the cap rock (Bachu *et al.*, 2007).

2.1.2. Residual gas (capillary) trapping mechanisms. Residual gas trapping takes place in the pore space at irreducible gas saturation, where carbon dioxide is immobile because of the interfacial tension between carbon dioxide and formation water, and flow is not possible even if a pathway is available. Residual gas trapping is based on disconnected gas bubbles being left in the wake of a migrating stream or plume of carbon dioxide (non-wetting phase) when water/brine (wetting phase) moves back into the pore space during an imbibition cycle, where brine displaces carbon dioxide (absorption of a gas by solid) after it has been expelled from the pore space during the drainage cycle by the injected and/or migrating carbon dioxide (Bachu, 2008). It is due to the hysteretic properties of relative permeability that, in multi-phase flow, results from irreversibility or path dependence of the relative permeabilities and capillary pressures on the saturation path and the saturation history (Juanes *et al.*, 2006).

The term 'imbibition cycle' is defined as the return of a wetting fluid, most commonly water, to a pore space when a non-wetting fluid (e.g. oil or carbon dioxide) retreats or moves out from it. An imbibition cycle follows a drainage cycle, defined as the displacement of a wetting fluid from some of the pore space by a non-wetting fluid. Imbibition-like process and drainage-like process are summarised in Table 3. Three main pore-scale displacement mechanisms have been presented in detail by several authors (Al-Futaisi and Patzek, 2003; Lenormand *et al.*, 1983; Valvatne and Blunt, 2004): piston-type displacement, cooperative pore-body filling and snap-off.

During the injection of carbon dioxide into geological formations, the non-wetting carbon dioxide displaces the wetting brine in a drainage-like process, and carbon dioxide saturation increases. Vertical and lateral flow paths are created

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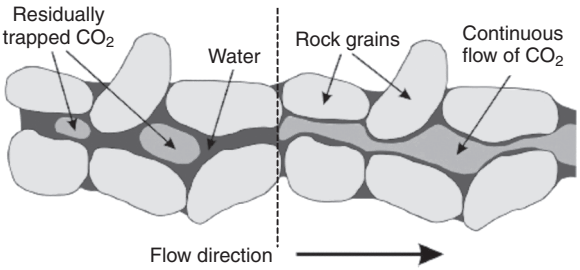
	Imbibition-like process	Drainage-like process
Edge of the plume	Trailing	Leading
CO_2 saturation	Decreases	Increases
Water saturation	Increases	Decreases
CO_2	Disconnected, trapped because of capillary pressure	Continuous connected cluster
Displacement	Piston-type displacement Cooperative body filling Snap-off	Piston-type displacement
Takes place	After injection	During and after injection
Schematic		

Table 3. Residual trapping of CO_2 (modified from CO2CRC, 2008)

as carbon dioxide migrates laterally away from the injection wells and to the top of the aquifer due to buoyancy forces. This is a drainage process in which the only mechanism for displacement of brine by carbon dioxide is piston-type displacement, where the carbon dioxide invades the porous medium in the form of a continuous, connected cluster (Juanes *et al.*, 2006).

Once injection stops, the buoyant carbon dioxide continues to migrate upward and laterally. At the leading edge of the plume, carbon dioxide displaces water in a drainage process (increasing gas saturation) while at the trailing edge of the plume, water displaces carbon dioxide in an imbibition-like process, increasing water saturation and gradually decreasing carbon dioxide saturation as the plume migrates. At the pore-scale, carbon dioxide is continuous in porous rocks. However, with a decrease in carbon dioxide saturation, the continuous carbon dioxide is disconnected at the points of pore constriction. Once a portion of the non-wetting carbon dioxide is separated from the continuum, in the form of blobs or ganglia, becoming effectively immobile, it is trapped because of capillary pressure (Juanes *et al.*, 2006). A carbon dioxide bubble that occupies a large pore space cannot pass through a pore neck because of interfacial tension (Suekane *et al.*, 2008). During this process, there are several physical mechanisms by which the water can displace the carbon dioxide (Juanes *et al.*, 2006). In addition to piston-type displacement, cooperative pore-body filling and snap-off may occur.

For water-wet rocks and a capillary-dominated flow regime, snap-off is the dominant mechanism at the pore-scale. The snap-off mechanism implies successive steps of liquid accumulation at pore throats, rearrangement of collars of the wetting liquid at throats into pore-spanning liquid lenses and further displacement of lenses or lamellae from pore throats. Capillary trapping of the non-wetting gas phase occurs during water flooding when the gas saturation is decreasing, and water saturation increases as it invades the pore space. The important point is that snap-off and cooperative filling may lead to disconnection and bypassing of the carbon dioxide (Kovscek and Radke, 2003). This leads to disconnection of the once-continuous plume into blobs and ganglia, which are effectively immobile. A trail of residual, immobile carbon dioxide is left behind the plume as it migrates upward (Juanes *et al.*, 2006). Over time, the residually trapped carbon dioxide dissolves into the formation water (Flett *et al.*, 2005).

Thus, residual gas trapping largely, if not entirely, takes place after injection has stopped, and cannot happen in structural and stratigraphic traps where only water drainage occurs when carbon dioxide is injected. Unlike in the case of structural and stratigraphic traps, the storage volume in residual gas trapping is time-dependent, increasing over time as the plume of carbon dioxide first spreads and then migrates (Bachu *et al.*, 2007).

Juanes *et al.* (2006) found that high injection rates result in more effective sequestration of carbon dioxide. A shorter injection period leaves less time for the buoyant carbon dioxide to reach the top of the formation, from which it is difficult to immobilise.

Recent simulation studies have highlighted the importance of capillary trapping mechanisms. There is one well-established method by which a non-wetting phase is rendered immobile in a short timescale (years to decades), namely water flooding. Qi *et al.* (2007) proposed a carbon storage strategy where carbon dioxide and brine are injected into an aquifer together (to improve the storage efficiency of the injection process) followed by chase brine injection alone, which rapidly traps the injected carbon dioxide. This process renders 80–95% of the carbon dioxide immobile in pore-scale droplets in the porous rock (Qi *et al.*, 2007). Simulation studies imply that only around 2% of the pore space will contain carbon dioxide if it is injected alone (Obi and Blunt, 2006). The carbon dioxide is trapped during the decades-long lifetime of the injection phase, reducing the need for extensive monitoring for centuries. This design has been demonstrated through the incorporation of a recent model of

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trapping and relative permeability hysteresis based on pore-scale modelling into a field-scale streamline based simulator (Qi *et al.*, 2007).

Modelling studies suggest that more than 60% of carbon dioxide may be trapped by residual carbon dioxide trapping by the end of the injection phase (100% after 1000 years), although laboratory experiments suggest somewhat lower percentages (IPCC, 2005).

2.2. Geochemical trapping mechanisms in saline formations

Carbon dioxide in the subsurface can undergo a sequence of geochemical interactions with the host rock and formation water that will further increase storage capacity and effectiveness (IPCC, 2005).

2.2.1. Solubility trapping. Over time, carbon dioxide that is not immobilised by residual trapping can dissolve in the formation water in a process called solubility trapping. The primary benefit of solubility trapping is that once carbon dioxide is dissolved, it no longer exists as a separate phase, thereby eliminating the buoyant forces that drive it upwards (IPCC, 2005).

First, carbon dioxide gas dissolves in an aqueous phase to produce weak carbonic acid (H_2CO_3) (Equations 1 and 2) (Xu *et al.*, 2004). The latter alters the pH (increases the acidity) of the groundwater and increases the solubilities of many minerals in the host rock matrix (Hitchon *et al.*, 1999; Xu *et al.*, 2004). This is followed by rapid dissociation of carbonic acid to form bicarbonate (Equation 3) and carbonate ions (Equation 4) as the rock dissolves (Xu *et al.*, 2004); this is accompanied by a rise in pH. Solubility trapping can be represented by the following reactions (Bachu, 2008; Xu *et al.*, 2004).

1 $$CO_{2(g)} \rightarrow CO_{2(aq)}$$

2 $$CO_{2(aq)} + H_2O \rightarrow H_2CO_{3(aq)}$$

3 $$H_2CO_{3(aq)} \rightarrow HCO^-_{3(aq)} + H^+$$

4 $$HCO^-_{3(aq)} \rightarrow CO^-_{3(aq)} + H^+$$

The pH of the brine significantly affects the solubility. Since dissolved carbon dioxide, $CO_{2(aq)}$ and H_2CO_3 (carbonic acid) dominate at low pH, the solubility of carbonate increases as the pH decreases (Soong *et al.*, 2003).

The solubility of carbon dioxide in water decreases with increasing temperature and increases with increasing pressure. The solubility of carbon dioxide in water also decreases with increasing water salinity by as much as one order of magnitude (Bachu, 2008; Holloway, 2005).

Solubility trapping is a continuous, relatively slow, time-dependent process that is predicted to be mostly active after cessation of injection (Bachu, 2008). With time, more carbon dioxide is dissolved in the formation water. The success of solubility trapping is aided by slow flow velocities of the formation water, as these will lead to a long residence time and allow more time for carbon dioxide to dissolve into the formation water (CO2CRC, 2008).

The rate of dissolution depends on how well the carbon dioxide mixes with the formation water once it is injected into the reservoir (Holloway, 2005). Carbon dioxide may mix with, and then dissolve in, formation water through three different processes: diffusion (an extremely slow process), dispersion and convection (Bachu, 2008). The rate at which solubility trapping occurs depends principally on the amount of free-phase carbon dioxide coming into contact with formation water unsaturated with carbon dioxide. Dissolution is rapid when formation water and carbon dioxide share the same pore space (IPCC, 2005). Once migration of the carbon dioxide has stopped (effectively shutting off dispersion) the water in contact with carbon dioxide becomes saturated with carbon dioxide; this slows down dissolution and diffusion becomes the main mixing process. This means that dissolution could be very slow (of the order of a few thousand years) unless there is some form of active mixing induced by fluid flow within the aquifer or convective mixing where the carbon-dioxide-saturated water is continuously replaced with unsaturated water (Bachu, 2008; Holloway, 2005). Because the density of carbon-dioxide-saturated water is around 1% greater than unsaturated formation water, if the rock permeability is sufficiently high and the aquifer is sufficiently thick, convective mixing within the pore system may occur. When the layer of saturated brine becomes thick enough (due to diffusion), density instability occurs and the plume of brine saturated with carbon dioxide migrates downwards, slowly diluting as it goes. The importance of convection is that it is typically orders of magnitude faster than diffusion at mixing the dissolved gas, and so it enhances the overall dissolution process (Bachu, 2008; Ennis-King and Paterson, 2002; Ghanbari *et al.*, 2006; Holloway, 2005; IPCC, 2005).

2.2.2. Mineral trapping. Once dissolved, carbon dioxide, in the form of bicarbonate or carbonate ions, can react directly or indirectly with the surrounding minerals, in the form of dissolved cations such as calcium, magnesium or iron, leading to precipitation of secondary carbonates (Lagneau *et al.*, 2005; Xu *et al.*, 2000, 2004). This so-called mineral trapping is the most permanent form of geological storage because it could immobilise carbon dioxide for very long periods of time (Xu *et al.*, 2004). Mineral trapping is believed to be comparatively slow, potentially taking hundreds to thousands of years or longer (IPCC, 2005).

The most basic chemical reactions that lead to mineral trapping are presented in Equations 5–8, in which calcium, magnesium, iron or other divalent ions react with carbonate ions to form minerals such as calcite, magnesite, siderite and dolomite, respectively (Bachu, 2008; Soong *et al.*, 2003; Xu *et al.*, 2004)

5 $$CO^{2-}_{3(aq)} + Ca^{2+} \rightarrow CaCO_{3(s)}\ (\text{calcite})$$

6 $$CO^{2-}_{3(aq)} + Mg^{2+} = MgCO_{3(s)}\ (\text{magnesite})$$

7 $$CO^{2-}_{3(aq)} + Fe^{2+} = FeCO_{3(s)}\ (\text{siderite})$$

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8	$CO_{3(aq)}^{2-} + Ca^{2+} + Mg^{2+} = CaMg(CO_3)_{2(s)}$ (dolomite)

For the more complex minerals commonly found in aquifers, the reaction is of the form

9	feldspars + clays + CO_2 = kaolinite + calcite + dolomite + siderite + quartz

where the carbon dioxide is permanently fixed as the carbonate minerals calcite, dolomite and siderite. Carbon dioxide mineral traps are most effective when the aquifer contains minerals that are proton sinks, that is, the basic silicate minerals such as feldspars and clay minerals. Consequently, mineral trapping of carbon dioxide is favoured in aquifers containing an abundance of clay minerals. Typically, siliciclastic (sandstone) aquifers are favoured over carbonate aquifers (Hitchon *et al*., 1999).

In a similar way as the pH of the brine affects solubility, it also affects precipitation of secondary carbonate minerals. Since HCO_3^- (bicarbonate) and CO_3^{2-} (carbonate) dominate at mid and high pH respectively, carbonate precipitation is favoured under more basic conditions and solubility is favoured under acidic conditions (Soong *et al*., 2003).

Like solubility trapping, mineral trapping is very dependent on the chemical composition of formation waters and of the rock matrix, and on temperature and pressure. In addition, it depends on the contact surface between the mineral grains and the formation water containing dissolved carbon dioxide, and on the flow rate of fluids past the interface. The flow rate depends on rock permeability, hydraulic gradients and water viscosity, which itself depends on water temperature and salinity, and much less so on pressure (IPCC, 2005).

Several authors have reported experimental data at a laboratory scale with the aim of better understanding the geochemistry of carbon dioxide storage in deep saline formations. Assessment of rock/water/carbon dioxide interactions under a range of temperatures from 100 to 350°C (which constitutes the range of reservoir temperatures common to many active hydrothermal systems) using two different types of rocks (granite and sandstone) indicated that the solution in acid conditions increases the dissolution rate of the rock, which will further lead the solution to become supersaturated and further the deposition of secondary minerals (Liu *et al*., 2003; Suto *et al*., 2001). Glauconite (iron-rich clay) and anorthite (calcium-rich plagioclase feldspar) were also tested for 30 days as potential host rocks for carbon dioxide injection. Surface spectroscopic techniques (e.g. X-ray photoelectron spectroscopy) indicated that elemental composition at mineral surfaces was modified as a result of the experiments, showing progress towards carbonate precipitation. Moreover, those changes agreed with equilibrium modelling calculations (Sass *et al*., 2001).

A system comprising arkose, shale and a synthetic brine was reacted at 200°C (selected to enhanced kinetic rates of silicate reactions) and 200 bar (chosen for consistency with and accessibility for sequestration scenarios) for 59 days to approach the steady state and then injected with carbon dioxide and allowed to react for a further 80 days. Upon conclusion of the experiment, magnesite precipitation was observed as well as textures (etch pits, mineralisation) indicating significant reaction of silicate minerals in the system (Kaszuba *et al*., 2003).

Several authors (Bruant *et al*., 2002; Giammar *et al*., 2005) have studied the high-pressure–high-temperature interaction between supercritical carbon dioxide, brine and forsteritic olivine (Mg_2SiO_4). They agreed that increased dissolution occurs with both increasing reaction temperature and pressure. Giammar *et al*. (2005) also identified the rate of nucleation as a barrier for magnesite precipitation. Thus, when solution compositions were manipulated in order to simulate conditions resulting from extensive forsterite dissolution, magnesite precipitation occurred. When forsterite is present, magnesite nucleation appeared to occur heterogeneously on the surface of forsterite particles but the presence of forsterite did not significantly accelerate magnesite nucleation relative to solid-free systems.

Xu *et al*. (2000, 2004) performed numerical modelling of geochemical processes using the Toughreact code (Xu *et al*., 2004) on three different aquifer mineralogies (glauconitic sandstone, Gulf Coast sediments and dunite) in the presence of carbon dioxide at 260 bar pressure. The mineral composition of the glauconite sandstone and Gulf Coast sediments consisted mainly of quartz and minor amounts of several carbonates and silicates; dunite is a forsterite (Mg_2SiO_4) rock with a small amount of fayalite (Fe_2SiO_4). The results indicated that carbon dioxide sequestration by matrix minerals varies considerably with rock type. Dunite appears to have the largest storage capacity (100 kg per m^3 of medium), but is rarely found in crustal rocks. The storage capacity of glauconitic sandstone and Gulf Coast sediments was reported as 17 and 90 kg per m^3 of medium, respectively. The trapping capacity depended in a very significant way on the mineral composition. For the glauconitic sandstone, trapping capacity was proportional to the abundance of glauconite (silicate-type mineral) present in the rock; similarly, for the Gulf Coast sediments, the capacity was proportional to the abundance of clinochlore-14A and daphnite-14 (silicate-type minerals). All three examples demonstrated that accumulation of carbonates in the rock matrix leads to a considerable decrease in porosity. This in turn adversely affects permeability, fluid flow in the aquifer and carbon dioxide injectivity (Xu *et al*., 2000, 2004). However, a potential positive side effect may be the self-sealing of cracks by the accumulation of these carbonates (Pearce *et al*., 2004).

Smith *et al*. (2004a, 2004b) conducted a preliminary assessment of the carbon dioxide sequestration potential of deep basalt hosted aquifers in Southern Idaho using the Geochemist Workbench computer code (Bethke, 2002). Mineral trapping seemed to become more important than solubility trapping after about 140 years, and more important than hydrodynamic trapping after about 285 years. In addition, the potential of basalt as an impermeable cap rock was also considered due to a significant decrease in porosity. However, it appears that there was minimal porosity reduction during the injection period, suggesting that mineralisation reactions may not reduce permeability during the operational lifetime of the injection well.

Rosenbauer *et al*. (2005) carried out carbon-dioxide–brine–rock experiments at 25 and 120°C from 100 to 600 bar and up to

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80 days reaction time. Their experiments reacted supercritical carbon dioxide with natural (shallow groundwater brine from Paradox Valley, Colorado) and synthetic brines (synthetic solution of a deep-aquifer brine also from Paradox Valley and a predominantly NaCl solution equivalent in ionic strength to the natural brine) in the presence and absence of limestone (carbonate rock) and plagioclase-rich arkose sandstone (silicate rock). Their work indicated that the solubility of carbon dioxide is enhanced in the presence of both rocks but, in limestone formations, storage of carbon dioxide is limited to ionic and hydrodynamic trapping.

Bateman *et al.* (2005) described a long-term laboratory experiment that reacted known quantities of minerals (a prepared mixture based on that of Utsira sand and containing mainly quartz (80%) and minor amounts of labradorite, k-feldspar, albite, calcite, dolomite, muscovite and chlorite) with carbon-dioxide-rich fluid aiming to use the experimental results as a test case with which to help validate predictive geochemical computer models. They reported a clear overestimation of the degree of reaction according to model predictions compared with experimental results. These differences are thought to be a reflection of the data quality (equilibrium and kinetic constants) used for the predictions rather than failure of the models themselves.

The effect of gas impurities derived from the carbon dioxide gas stream in the rock/water/gas interactions upon co-injection deep underground is attracting much interest because there might be cost savings in the co-storage of carbon dioxide and other contaminants (the carbon dioxide capture process is by far the most expensive part of the whole CSS system) (Garcia *et al.*, 2008; Jacquemet *et al.*, 2005; Palandri and Kharaka, 2005; Palandri *et al.*, 2005; Xu *et al.*, 2007). Acid gas injection experience can provide some insights into the process, and recent modelling and experimental studies targeting ferric iron-bearing sediments as host rocks show very promising results, in terms of successful mineral trapping, after co-injection of mixtures of carbon dioxide and sulfur dioxide (Garcia *et al.*, 2008). This concept could also be applied to a new range of technologies currently under study to develop integrated pollutant removal systems for coal combustion (Chiesa *et al.*, 2005; IPCC, 2005). Clarity is needed – from both a technical and legal perspective – about the concentration of contaminants that could be co-injected and stored with carbon dioxide.

2.2.3. Hydrodynamic trapping. Hydrodynamic trapping can occur in saline formations that do not have a closed trap, but where fluids migrate very slowly over long distances. When carbon dioxide is injected into a formation, it displaces saline formation water and then migrates upwards because it is less dense than water. When it reaches the top of the formation, it continues to migrate as a separate phase until it is trapped as residual carbon dioxide or, in local structural or stratigraphic traps, within the sealing formation.

In the longer term, significant quantities of carbon dioxide appear to dissolve in the formation water and then migrate with the groundwater. Where the distance from the deep injection site to the end of the overlying impermeable formation is hundreds of kilometres, the timescale for the fluid to reach the surface from the deep basin could be millions of years (IPCC, 2005).

When carbon dioxide, or any fluid, is injected into an aquifer at an appropriate pressure below the fracture pressure, it moves out from the injection well and flows with the natural flow regime. Because strata in most sedimentary basins are not horizontal, the possible injection aquifers will vary from place to place, and no aquifer is likely to be a potential injection aquifer across an entire basin. Once outside the injection-well radius of influence, the flow of immiscible carbon dioxide will travel at the same speed as the regional flow system. For the Alberta Basin, simple calculations indicate that after times of the order of millions of years the carbon dioxide may still remain within tens of kilometres of the injection site. The water velocity may be even lower than noted above because of down-dip buoyancy-driven flow caused by the increased salinity of formation waters with depth in the south-west part of the basin, opposing the up-dip topographically driven flow to the north-east. This geological timescale trapping of carbon dioxide in deep regional aquifers as a result of very low flow velocity is called hydrodynamic trapping because it depends on the hydrodynamic regime of the formation waters (Hitchon *et al.*, 1999).

3. MATCHING CARBON DIOXIDE SOURCES WITH SEDIMENTARY BASINS (SOURCE–SINK MATCHING)

Carbon dioxide emissions arise from a different number of sources, such as power generation from fossil fuel combustion, industrial, residential and transport sectors. Emissions from the transport and residential sectors are generally very widely distributed geographically with a large number of emission sources but small carbon dioxide emissions from each (i.e. diffuse). They are also mobile (in the case of the transport sector), making them unsuitable for capture at present. The other two sectors, power generation and industrial, have large emission volumes and therefore allow for the easier capture of carbon dioxide. Over 60% of global carbon dioxide emissions come from the power and industry sectors. The Intergovernmental Panel on Climate Change (IPCC, 2005) has identified over 7500 large (above 0·1 $MtCO_2$/year) carbon dioxide emission sources. The majority of these have concentrations of carbon dioxide that are typically lower than 15% by volume and are concentrated in four main different areas around the world: North America, north-west Europe, South East Asia and southern Asia. These regions account for over 90% of the emissions. Although North America is the region with the largest number of stationary sources (37%), followed by Asia (24%) and Europe (14%), the IEA Greenhouse Gas Programme (IEAGHG, 2002) pointed out that the largest emitters of carbon dioxide from stationary sources are Asia (41%), North America (20%) and Europe (13%).

Opportunities for CSS in a specific region will require a source–sink matching evaluation. The best candidates for carbon dioxide storage have to be selected on not only their technical suitability but also on an assessment of economic, safety and environmental aspects. The latter could hamper the development of a site storage potential even if its capacity and properties are adequate for carbon dioxide injection (IPCC, 2005; Winter and Bergman, 1993). Bradshaw and Dance (2004) mapped regional emission clusters for 12 regions across the world, along with the available storage opportunities (sedimentary basins) within each region. A radius of 300 km around the emission sources was chosen as an indication of

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likely transport distances for potential source-to-storage matches. At a global scale, some potential areas for the geological storage of carbon dioxide could be identified, such as those with a high number of emission sites but only few opportunities for geological storage (e.g. China and the Indian subcontinent). Conversely, as in South America, there are other regions where the storage capacity was found to be much higher than the number of emission sites. Due to several assumptions made and the general character of the map, a more detailed and comprehensive geological assessment needs to be carried out on a regional and local site-specific basis in order to ascertain the quality and viability of each of the sedimentary basins and specific proposed emission sites.

Bradshaw *et al.* (2004) completed a more specific analysis of the potential for the geological storage of carbon dioxide in Australia. An emissions map was generated identifying 35 potential stationary point sources that represented 90% of the emissions that could be sequestered. Their classification into nodes aimed to reduce infrastructure costs of establishing a source-to-site injection system. Storage capacity was also estimated for 65 potential sites for carbon dioxide injection. A play and prospect type assessment (where five risk factors were considered) was used to calculate a potential risk storage capacity of 720 $GtCO_2$ or 1600 years' of Australia's annual emissions. Considering economics, nodes with large emission volumes and those within 300 km of a technically viable site, Australia has a more realistic storage potential of between 100 and 115 $MtCO_2$/year, which is around 25% of its annual emissions.

In the European scenario (25 member states), the Chalmers carbon dioxide storage database (Kjarstad and Johnsson, 2007) registered 820 different reservoirs with a technical trapped storage potential between 115 and 135 $GtCO_2$, roughly distributed among onshore and offshore sites, that could potentially reduce carbon dioxide emissions from stationary sources in Europe for over 70 years. Storage in aquifers accounted for between 64 and 82% of that total storage potential. Five different countries within the EU were responsible for the majority of this storage potential – Denmark, Germany, The Netherlands, Norway and the UK – accounting for almost 95% of the identified storage sites within Europe.

The Chalmers database also describes European power plant structure and contains all thermal and hydropower plants with a power output of at least 10 MW and all offshore wind farms and cross-border transmission lines. Almost all the information in the database was collected through direct contact with each utility. The database identified 522 GW of total net capacity from thermal plants in operation within EU member states. A total capacity of 170 GW is registered as under construction or planned; 50% of the units are gas-fuelled, following a clear shift in European power plant infrastructure to natural-gas-fuelled power plants over the last decade.

Due to the large proportion of old coal plants and many oil fields close to depletion, the most promising storage option in the UK in a near-term scenario might be EOR, with storage potential estimates of around 700 $MtCO_2$ over 20 years. If the EOR potential was utilised by 2031, and bearing in mind the UK potential for carbon dioxide storage in offshore aquifers might be 20 Gt out of the 30 Gt offshore storage capacity in subsurface reservoirs, it seems likely that the main storage option after 2031 would be storage in either gas fields or near-shore aquifers. The most significant formation is the Bunter sandstone, which is just off the coast in the southern North Sea basin and has an estimated capacity of 14·3 Gt for carbon dioxide storage. Although the number of oil and gas fields is much larger than that of aquifers, the capacity is significantly higher in the latter and it is clear that the UK's main potential storage capacity lies offshore.

A key point for full deployment of an integrated CCS system will be the transportation of carbon dioxide from sources to storage sites. Compressed gas occupies less volume and can be transported by pipeline; carbon dioxide transportation by pipeline is not a new concept and has been practised for over 25 years, extending over more than 2500 km in the western USA, mainly to supply carbon dioxide from natural sources to EOR projects in west Texas and elsewhere. Cortez, Bravo, Sheep Mountain and Weyburn are good examples of long-distance carbon dioxide pipelines, with capacities ranging between 5 and 19·3 $MtCO_2$/year and covering lengths of 328–808 km (IPCC, 2005). The economics of the technological system connecting carbon dioxide point sources and geologic storage reservoirs are another key issue to consider in a potential CCS scenario. Pipeline costs depend on several factors – terrain, operating pressures and temperatures, presence of impurities in the carbon dioxide stream, possible compression stations, etc. While source costs represent the majority of the costs in an integrated CCS system, the shape of the average cost curve is determined by variability of the source and reservoir costs and capacities (Bielicki, 2008).

4. ECONOMICS OF CARBON DIOXIDE STORAGE IN SALINE FORMATIONS

This section considers one of the three CCS component costs – storage cost. Costs for the various components of a CCS system vary widely, depending on the reference plant (whether coal- or gas-fired, hydrogen and ammonia production, gas processing or other industrial sources) and the wide range of carbon dioxide source, transport and storage situations. The cost of employing a full CCS system is dominated by the cost of capture, which is 15–75 US$/$tCO_2$ net captured for a coal- or gas-fired power plant. Typical costs for transportation are in the range 1–8 US$/$tCO_2$ transported; geological storage ranges from 0·5 to 8 US$/$tCO_2$ net injected, plus 0·1–0·3 US$/$tCO_2$ injected for monitoring and verification (see Table 4).

Over the next decade, the cost of capture could be reduced by 20–30% and more should be achievable by new technologies that are still in the research or demonstration phase and also by economies of scale (IPCC, 2005).

The equipment and technologies required for storage are already widely used in the energy industry, so that costs can be estimated with confidence (IPCC, 2005). However, there will be a significant range and variability in costs due to site-specific factors such as

(*a*) type of storage option
(*b*) onshore versus offshore
(*c*) reservoir and injection parameters (permeability, thickness, pressure, depth)

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Option	Representative cost	Remarks
Capture from a coal- or gas-fired power plant	15–75US$/$tCO_2$ net captured	Net costs of captured CO_2 compared with the same plant without capture
Transportation	1–8US$/$tCO_2$ transported	Per 250 km pipeline or shipping for mass flow rates of 5 (high end) to 40 (low end) $MtCO_2$/year
Geological storage	0·5–8US$/$tCO_2$ net injected	Excluding potential revenues from EOR or ECBM
Geological storage: monitoring and verification	0·1–0·3US$/$tCO_2$ injected	This covers pre-injection, injection and post-injection monitoring, and depends on the regulatory requirements

Table 4. Estimates of CCS costs (IPCC, 2005)

(*d*) carbon dioxide purity (Allinson *et al.*, 2003; Bock *et al.*, 2003; IPCC, 2005).

The storage cost varies for different storage options. The IPCC (2005) presented a compilation of three comprehensive assessments of carbon dioxide storage cost estimates for the USA, Australia and Europe. Table 5 presents estimates from those three studies and shows that representative estimates of the cost for storage in saline formations typically fall between 0·5 and 7·7 US$/$tCO_2$ stored. Offshore storage costs are significantly higher than those for a reservoir in an onshore location (3·4–7·7 US$/$tCO_2$ stored), due to the need for platforms or sub-sea facilities and higher offshore drilling costs (Allinson *et al.*, 2003; Bock *et al.*, 2003; IPCC, 2005). Storage in saline formations does not have any economic benefits, which makes it less economically attractive than EOR, enhanced gas recovery (EGR) or ECBM production.

Similar estimates were presented for disused oil and gas reservoirs, ranging from 1·7 US$/$tCO_2$ stored onshore to 6 US$/$tCO_2$ stored in offshore sites. The costs depend on the depth of the reservoir and the reuse of platforms. Disused fields may benefit from reduced exploration and monitoring costs. The costs of geological storage of carbon dioxide may be offset by additional revenues for production of oil or gas, where carbon dioxide injection and storage is combined with enhanced oil or gas recovery or ECBM. At present, in commercial EOR and ECBM projects that use carbon dioxide injection, the carbon dioxide is purchased for the project and is a significant proportion of operating costs (68% of the total costs for a number of EOR fields in Texas); this indicates the scale of benefit for EOR to offset carbon dioxide storage costs. The indicative value of the potential benefit from enhanced oil production as a result of carbon dioxide storage is usually in the range 0–16 US$/$tCO_2$ stored. In some cases, there is no benefit from EOR. In general, higher benefits will occur at high-oil-price scenarios and for highly favourable sites. Cost estimates show that CO_2-EGR can provide a benefit of 4–16 US$/$tCO_2$, depending on the price of gas and the effectiveness of recovery (IPCC, 2005). The IEAGHG programme developed a global cost curve for CO_2-ECBM, with storage costs ranging from −20 to +150 US$/$tCO_2$. The revenue benefit of CO_2-ECBM will depend on gas prices and on well cost because many wells are required (IPCC, 2005).

Storage option type, depth and geological characteristics affect the number, spacing and cost of wells (which are the major cost component), as well as the facilities cost (Bock *et al.*, 2003; IPCC, 2005). The design basis for each of the geological and injection parameters for the aquifer storage option and the calculated number of wells are given in Table 6 (Bock *et al.*, 2003). An increase in pressure and depth causes an increase in storage costs, in particular well and compression costs (Bock *et al.*, 2003; IPCC, 2005). A higher reservoir pressure results in lower injectivity (not desired) but higher carbon dioxide densities (desired), and therefore a moderate pressure is optimal. In general, the permeability is the most critical parameter in determining costs. Reservoirs that are thick, shallow and have high permeability require a smaller number of wells and, therefore, present a lower storage cost (Bock *et al.*, 2003).

Other gases, such as methane, hydrogen sulfide, nitrogen, nitrous oxide and oxides of sulfur, could be potentially captured together with carbon dioxide. However, although capture costs may be reduced, it is more costly to inject and store other gases with carbon dioxide because of their corrosive and hazardous nature. Reduced capture costs would be offset by an increase in the costs of storage because, for the same mass of carbon

On or offshore	Location	US$/$tCO_2$ stored			Reference
		Low	Mid	High	
Onshore	Australia	0·2	0·5	5·1	Allinson *et al.* (2003)
Onshore	Europe	1·9	2·8	6·2	Hendricks *et al.* (2002)
Onshore	USA	0·4	0·5	4·5	Bock *et al.* (2003)
Offshore	Australia	0·5	3·4	30·2	Allinson *et al.* (2003)
Offshore	North Sea	4·7	7·7	12·0	Hendricks *et al.* (2002)

Table 5. Compilation of CO_2 storage cost estimates in saline formations (including capital, operating and site characterisation costs; excluding monitoring costs, remediation and any additional costs required to address long-term liabilities) (modified from IPCC, 2005)

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Parameter	Aquifer base case	Aquifer high cost case	Aquifer low cost case
Pressure: MPa	8·4	11·8	5·0
Thickness: m	171	42	703
Depth: m	1239	1784	694
Permeability: md (millidarcy)	22	0·8	585
Injection rate per well*: t/day	9363	82	889 495
Number of wells*	1	91	1

* Calculated

Table 6. Design bases for aquifer storage option (Bock *et al*., 2003)

dioxide (tonnes), large gas volumes have to be handled. This requires higher compression power and/or bigger pipelines, possibly more wells and, in the offshore case, possibly larger platforms. For instance, storing a mixture of gases with 50% carbon dioxide content can increase the capital and operating costs of storage by about 50% (Allinson *et al*., 2003).

5. CONCLUSIONS

Carbon capture and storage (CCS) technologies are considered to be an essential bridge between present fossil energy and a future renewable/hydrogen-based energy, as they have the potential to reduce overall mitigation costs and increase flexibility in achieving a reduction in GHG emissions.

Carbon dioxide storage in saline formations offers a much higher storage capacity, the global value of which is estimated to be 1000–56 000 $GtCO_2$ (depending on the assessment) more than other more economically attractive options such as EOR and ECBM. In a European scenario, storage in saline aquifers might account for between 64 and 82% of the total storage potential. For the UK, the storage capacity lies mainly offshore and EOR appears to be the most promising option in a near-term scenario. However, it seems likely this option will be replaced by storage in gas fields or near-shore aquifers after 2031, due to the large proportion of old coal plants and oil fields close to depletion. Research into different areas covering geochemical, engineering and economic aspects is needed for precise and reliable storage estimates, alongside better understanding of the carbon dioxide underground trapping mechanisms and the interplay between them as a key point to integrate the different methods to estimate the carbon dioxide storage potential in saline aquifers.

On the economics side, the storage cost varies for different storage options. Storage in saline formations is less attractive than others (EOR, EGR and ECBM). Representative estimates of the cost for storage in saline formations typically falls between 0·5 and 7·7 US$/$tCO_2$ stored, with higher costs for offshore storage options compared with onshore facilities.

One of the key issues to enable implementation of the technology is to ascertain whether the risks of leakage from a storage site and any resulting adverse impact on humans and the natural environment are acceptable. The extensive use and production of carbon dioxide has provided well-established engineering and procedural controls for dealing with the hazards of cryogenic and compressed carbon dioxide, and generally show that those risks could be easily manageable. A more challenging task is to understand and predict carbon dioxide releases and their characteristics in a given geographic or geologic setting. Therefore, a thorough site characterisation will be required in every risk assessment procedure.

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REFERENCES

Al-Futaisi A and Patzek TW (2003) The impact of wettability alteration on two-phase flow characteristics of sandstones: a quasi-static description. *Water Resources Research* **39(2)**: 1042–1054 doi:10.1029/2002WR001366.

Allinson WG, Nguyen DN and Bradshaw J (2003) The economics of geological storage of CO_2 in Australia. *Australian Petroleum Production and Exploration Association (APPEA) Journal* **43(1)**: 623–636.

Bachu S (2008) CO_2 storage in geological media: role, means, status and barriers to deployment. *Progress in Energy and Combustion Science* **34(2)**: 254–273.

Bachu S, Bonijoly D, Bradshaw J *et al.* (2007) CO_2 storage capacity estimation: methodology and gaps. *International Journal of Greenhouse Gas Control* **1(4)**: 430–443.

Bateman K, Turner G, Pearce JM *et al.* (2005) Large-scale column experiment: study of CO_2, porewater, rock reactions and model test case. *Oil & Gas Science and Technology* **60(1)**: 161–175.

Bethke CM (2002) *The Geochemist's Workbench: A User's Guide to Rxn, Act2, Tact, React, and Gtplot*, Release 4.0. University of Illinois, Urbana, IL.

Bickle M, Chadwick A, Huppert HE, Hallworth M and Lyle S (2007) Modelling carbon dioxide accumulation at Sleipner: implications for underground carbon storage. *Earth and Planetary Science Letters* **255(1–2)**: 164–176.

Bielicki J (2008) Returns to scale in the infrastructure for carbon capture and storage. *Proceedings of the 7th Annual Conference on Carbon Capture & Sequestration, Pittsburgh, PA*. US Department of Energy, NETL (National Energy Technology Laboratory) CDRom.

Bock B, Rhudy R, Herzog H *et al.* (2003) *Economic Evaluation of CO_2 Storage and Sink Enhancement Options*. TVA Public Power Institute, Alabama, USA.

Bradshaw J and Dance T (2004) Mapping geological storage prospectivity of CO_2 for the world's sedimentary basins and regional source to sink matching. *Proceedings of the 7th International Conference on Greenhouse Gas Technologies (GHGT-7), Vancouver*. Elsevier Ltd, Oxford, UK, pp. 583–591.

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Bradshaw J, Allinson G, Bradshaw BE *et al.* (2004) Australia's CO_2 geological storage potential and matching of emission sources to potential sinks. *Energy* **29(9–10)**: 1623–1631.

Bruant Jr RG, Giammar DE, Myneni SCB and Peters CA (2002) Effect of pressure, temperature, and aqueous carbon dioxide concentration on mineral weathering as applied to geologic storage of carbon dioxide. *Proceedings of the 6th International Conference on Greenhouse Gas Control Technologies (GHGT-6), Kyoto*. Elsevier Ltd, Oxford, UK, pp. 1609–1612.

Chiesa P, Consonni S, Kreutz T and Williams R (2005) Co-production of hydrogen, electricity and CO_2 from coal with commercially ready technology. Part A: performance and emissions. *International Journal of Hydrogen Energy* **30(7)**: 747–767.

Chilingar GV, Buryakovsky LA, Eremenko NA and Gorfunkel MV (2005) *Geology and Geochemistry of Oil and Gas*. Elsevier, New York.

Chiquet P, Daridon JL, Broseta D and Thibeau S (2007) CO_2/water interfacial tensions under pressure and temperature conditions of CO_2 geological storage. *Energy Conversion and Management* **48(3)**: 736–744.

CO2CRC (Cooperative Research Centre for Greenhouse Gas Technologies) (2008) *Storage Capacity Estimation, Site Selection and Characterisation for CO_2 Storage Projects*. CO2CRC, Canberra, Australia, Report RPT08-1001, pp. 1–60.

Ennis-King J and Paterson L (2002) Rate of dissolution due to convective mixing in the underground storage of carbon dioxide. *Proceedings of the 6th International Conference on Greenhouse Gas Control Technologies, Kyoto*. Elsevier Ltd, Oxford, UK, pp. 507–510.

Flett MA, Gurton RM and Taggart IJ (2005) Heterogeneous saline formations: long-term benefits for geosequestration of greenhouse gases. *Proceedings of the 7th International Conference on Greenhouse Gas Control Technologies, Vancouver*. Elsevier Ltd, Oxford, UK, pp. 501–509.

Garcia S, Rosenbauer RJ, Palandri JL and Maroto-Valer MM (2008) Sequestration of non-pure carbon dioxide streams in geological formations. *Proceedings of the 7th Annual Conference on Carbon Capture & Sequestration, Pittsburgh, PA*. US Department of Energy, NETL (National Energy Technology Laboratory), USA, CDRom.

Ghanbari S, Al-Zaabi Y, Pickup GE *et al.* (2006) Simulation of CO_2 storage in saline aquifers. *Chemical Engineering Research and Design* **84(9)**: 764–775.

Giammar DE, Bruant Jr RG and Peters CA (2005) Forsterite dissolution and magnesite precipitation at conditions relevant for deep saline aquifer storage and sequestration of carbon dioxide. *Chemical Geology* **217(3–4)**: 257–276.

Hendricks C, Graus W and van Bergen F (2002) *Global Carbon Dioxide Storage Potential and Costs*. Ecofys and the Netherland Institute of Applied Geoscience TNO, Ecofys Report EEP02002, 63pp.

Hitchon B, Gunter WD, Gentzis T and Bailey RT (1999) Sedimentary basins and greenhouse gases: a serendipitous association. *Energy Conversion and Management* **40(8)**: 825–843.

Holloway S (2005) Underground sequestration of carbon dioxide – a viable greenhouse gas mitigation option. *Energy* **30(11–12)**: 2318–2333.

IEAGHG (International Energy Agency Greenhouse Gas R&D Programme) (2002) *Building the Cost Curves for CO_2 Storage; Part 1: Sources of CO_2*. IEAGHG, Cheltenham, Gloucestershire, Report PH4/9, pp. 1–48.

IEAGHG (2010) www.ieagreen.org.uk (accessed 15/03/2010).

IPCC (Intergovernmental Panel on Climate Change) (2005) *IPCC Special Report on Carbon Dioxide Capture and Storage Prepared by Working Group III of Intergovernmental Panel on Climate Change*. Cambridge, Cambridge University Press, pp. 1–442.

Jacquemet N, Pironon J and Caroli E (2005) A new experimental procedure for simulation of H_2S + CO_2 geological storage. Application to well cement aging. *Oil & Gas Science and Technology* **60(1)**: 193–206.

Juanes R, Spiteri EJ, Orr Jr FM and Blunt MJ (2006) Impact of relative permeability hysteresis on geological CO_2 storage. *Water Resources Research* **42**: W12418 doi:10.1029/2005WR004806.

Kaszuba JP, Janecky DR and Snow MG (2003) Carbon dioxide reaction processes in a model brine aquifer at 200°C and 200 bars: implications for geologic sequestration of carbon. *Applied Geochemistry* **18(7)**: 1065–1080.

Kjarstad J and Johnsson F (2007) The European power plant infrastructure – presentation of the Chalmers energy infrastructure database with applications. *Energy Policy* **35(7)**: 3643–3664.

Kovscek AR and Radke CJ (2003) Pressure-driven capillary snap-off of gas bubbles at low wetting-liquid content. *Colloids and Surfaces A: Physicochemical and Engineering Aspects* **212(2–3)**: 99–108.

Laate M (2008) StatoilHydro's R&D on CO_2 storage and capture – 12 years of CO_2 storage. *Proceedings of the 7th Annual Conference on Carbon Capture & Sequestration, Pittsburgh, PA*. US Department of Energy, NETL (National Energy Technology Laboratory), USA, CDRom.

Lackner KS, Butt DP, Wendt CH and Ziock H (1998) *Mineral Carbonates as Carbon Dioxide Sinks*. Los Alamos National Laboratory, Los Alamos, USA, Report LA-UR-98-4530, pp. 1–10.

Lagneau V, Pipart A and Catalette H (2005) Reactive transport modelling of CO_2 sequestration in deep saline aquifers. *Oil & Gas Science and Technology* **60(2)**: 231–247.

Lenormand R, Zarcone C and Sarr A (1983) Mechanisms of the displacement of one fluid by another in a network of capillary ducts. *Journal of Fluid Mechanics* **135(1)**: 337–353.

Liu L, Suto Y, Bignall G, Yamasaki N and Hashida T (2003) CO_2 injection to granite and sandstone in experimental rock/hot water systems. *Energy Conversion and Management* **44(9)**: 1399–1410.

Mazzotti M, Pini R and Storti G (2009) Enhanced coalbed methane recovery. *Journal of Supercritical Fluids* **47(3)**: 619–627.

Obi EOI and Blunt MJ (2006) Streamline-based simulation of carbon dioxide storage in a North Sea aquifer. *Water Resources Research* **42**: W03414 doi: 10.1029/2004WR003347.

Palandri JL and Kharaka YK (2005) Ferric iron-bearing sediments as a mineral trap for CO_2 sequestration: iron reduction using sulfur-bearing waste gas. *Chemical Geology* **217(3–4)**: 351–364.

Palandri JL, Rosenbauer RJ and Kharaka YK (2005) Ferric iron in sediments as a novel CO_2 mineral trap: CO_2–SO_2 reaction with hematite. *Applied Geochemistry* **20(11)**: 2038–2048.

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Pearce J, Czernichowski-Lauriol I, Lombardi S *et al.* (2004) A review of natural CO_2 accumulations in Europe as analogues for geological sequestration. *Geological Society of London, Special Publications* **233(1)**: 29–41.

Qi R, Beraldo V, LaForce T and Blunt MJ (2007) Design of carbon dioxide storage in a North Sea aquifer using streamline-based simulation. *SPE Annual Technical Conference and Exhibition, Anaheim*, CA. SPE (Society of Petroleum Engineers), Texas, USA.

Rosenbauer RJ, Koksalan T and Palandri JL (2005) Experimental investigation of CO_2–brine–rock interactions at elevated temperature and pressure: implications for CO_2 sequestration in deep-saline aquifers. *Fuel Processing Technology* **86(14–15)**: 1581–1597.

Rutqvist J and Tsang C (2002) A study of caprock hydromechanical changes associated with CO_2-injection into a brine formation. *Environmental Geology* **42(2–3)**: 296–305.

Sass BM, Gupta N, Ickes JA *et al.* (2001) Interaction of rock minerals with carbon dioxide and brine: a hydrothermal investigation. *Proceedings of the 1st National Conference on Carbon Sequestration, Washington, DC*. NETL (National Energy Technology Laboratory), USA, CDRom.

Selley RC (1998) *Elements of Petroleum Geology*. Elsevier, New York.

Smith RW, McLing TL, Barrash W, Clement WP and Erickson NP (2004a) Geologic sequestration of CO_2: a uniform strategy for assessing mineralization trapping potential across rock types. *Proceedings of the 3rd Annual Conference on Carbon Capture and Sequestration, Alexandria, VA*.

Smith RW, McLing TL, Barrash W and Kaszuba J (2004b) *Carbon Sequestration Potential of Basins of Southern Idaho*. Big Sky Carbon Sequestration Partnership, Montana, USA.

Soong Y, Jones JR, Harrison DK *et al.* (2003) Mineral trapping of CO_2 with Oriskany brine. *Proceedings of the 2nd Annual Conference on Carbon Sequestration, Alexandria, VA*.

Suekane T, Nobuso T, Hirai S and Kiyota M (2008) Geological storage of carbon dioxide by residual gas and solubility trapping. *International Journal of Greenhouse Gas Control* **2(1)**: 58–64.

Suto Y, Liu L, Hashida T, Tsuchiya N and Yamasaki N (2001) Experimental study of rock/water/CO_2 interaction at temperatures of 100–350°C. *Proceedings of the 10th International Symposium on Water–Rock Interaction (WRI-10), Villasimius, Italy*. Taylor and Francis, London, pp. 341–344.

Valvatne PH and Blunt MJ (2004) Predictive pore-scale modeling of two-phase flow in mixed wet media. *Water Resources Research* **40**, doi: 10.1029/2003WR002627.

Winter EM and Bergman PD (1993) Availability of depleted oil and gas-reservoirs for disposal of carbon-dioxide in the United States. *Energy Conversion and Management* **34(9–11)**: 1177–1187.

Xu T, Apps JA and Pruess K (2000) *Analysis of Mineral Trapping for CO_2 Disposal in Deep Aquifers*. Lawrence Berkeley National Laboratory, Berkeley, CA, Report LBNL-46992, pp. 1–62.

Xu T, Apps JA and Pruess K (2004) Numerical simulation of CO_2 disposal by mineral trapping in deep aquifers. *Applied Geochemistry* **19(6)**: 917–936.

Xu T, Apps J, Pruess K and Yamamoto H (2007) Numerical modeling of injection and mineral trapping of CO_2 with H_2S and SO_2 in a sandstone formation. *Chemical Geology* **242(3–4)**: 319–346.

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